Nikhil Mundhra

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EDUCATION & SKILLS

BSc. in Computer Science with **Economics** & **Applied Math** | New York University Abu Dhabi — Aug 2023 – May 2027

- **Cumulative GPA:** 3.5 / 4.0 | **Focus Areas:** Machine Learning and AI Engineering
- Selected for and completed Study Away terms at **NYU CAS**, **NYU Tandon**, and **NYU Paris** — Jan 2025 – Dec 2025

UAE Top 3 Adaptive City Hackathon Award Winner | Organized by the Transpose Platform

Distinction Certificate | CEMC Waterloo CCC (Canadian Computing Competition)

Languages: Python | SQL | C++ | Bash | JavaScript | TypeScript | C | Java | R

AI & Machine Learning: PyTorch | TensorFlow | NumPy | Scikit-Learn | Pandas | LLMs | RAG | LangGraph | Time-Series

MLOps, Cloud & Systems: Linux | Docker & Kubernetes | CI/CD (Github & Azure DevOps) | Git | GCP | AWS | NYUAD HPC

Software & Backend Systems: PostgreSQL | Redis | Flask | REST APIs | Pytest

PROFESSIONAL EXPERIENCE

Machine Learning Researcher (Capstone Project) | NYU Abu Dhabi, Center for Brain and Health — May 2026 – Present

- Engineered 2.5D Bi-Planar ResNet, 3D U-Net, and Hybrid TransUNet models for RNFL segmentation (320×768×320) under *Dr. Bas Rokers*, achieving **0.953** Surface Dice and **4.71 µm** (~1.5-px) boundary error (beating commercial Solix/Heidelberg baselines with **+64.6%** audit gain and **47%** lower P95 tail error).
- Developed a hierarchical ConvNeXt V2 classifier with multi-scale CBAM attention and Grad-CAM across **88,800+** OCT B-scans (**7,536** patients), achieving **98%** triage F1 across **12** retinal pathologies under zero-leakage strict cross-validation.
- Architected end-to-end PyTorch, MONAI, and scikit-learn pipelines for automated multi-task training (Tversky, continuous boundary regression, and BMO cup losses), validation, and clinical reporting across 237 volumetric 3D scans (~75,800 slices), reducing per-volume inference latency by up to **8x** (7.1s vs. 45–60s heuristic baseline).
- Scaled distributed multi-GPU training on NYUAD's Jubail HPC cluster across NVIDIA H200, A100, and 4x V100 GPUs (128 GB VRAM) via PyTorch DDP under *Prof. Gabriel Falcão*, optimizing mixed-precision (BF16/FP16 with FP32 loss upcasting) at effective batch size 64 to reach **95%–98%** GPU SM utilization.
- Targeting publication at IPMI for a comparative study of ML architectures and their biomarker detection accuracy.

AI Intern | Kanerika Software Pvt. Ltd. – Hyderabad, India — Jun 2026 – Aug 2026

- Engineered an agentic fraud engine that synthesizes financial typologies (rent skimming, subsidy theft) into persistent SQL rules across a data lake, pairing deterministic flagging with LLM audit explainability.
- Developed a legal document compliance AI engine with a multi-tenant distributed Microsoft WOPI architecture and native Word add-in end-to-end, enabling seamless integration into client ecosystems.
- Conducting functional and technical REST API testing and debugging of Kanerika AI Labs' products (including Karl)

Cyber Security & Project Management Intern | Alstom – India — Jun 2025 – Aug 2025

- Conducted multi-site research of RS17 Trains, analyzing the DfQ V-cycle, gate reviews, and system V&V checkpoints.
- Shadowed development of OT cybersecurity models, including network segmentation and IEC 62443 compliance.

Marketing Analyst Intern | **Phool.Co – New Delhi, India** — Jun 2024 – Aug 2024

- Reported geographical, financial, and seasonal buying behaviors using multi-variable data models in STATA & R.

PROJECTS

Developer (Team Lead) | AeroLodge Metrics - Department of Culture & Tourism, UAE funded project — Aug 2026 – Oct 2026

- Facilitated flight routes & hotel capacity decisions for the DCT using proprietary data by curating a lake architecture, and modeling noisy multi-horizon time-series via Constrained Convolution Filters, outperforming deep neural baselines.

Developer (Team Lead) | Wealth Advisor - A Shenzhen FinTechathon project — Sep 2026 – Oct 2026

- Built a B2B2C Dynamic Expat Wealth Agent (DEWA) for autonomous AI wealth advisory for advisors and end-clients.

Founder | VivahGo Planners – An AI-driven SaaS platform for wedding workspaces — Mar 2026 – Present

- Handled Marketing, Operations, Software Development, Finances, and Legal compliances of founding a startup

Developer (Solo) | TakeOver – Vercel & GCP Deployed Full Stack AI Harness for WhatsApp — Jul 2026 – Sep 2026

- Engineered TakeOver agent pipeline to use semantic chat context, user guidelines, and smart permission polling

LEADERSHIP

Founder and **President** | Hindu Student Association (Peacock), NYU Abu Dhabi — Mar 2024 – Present

- Kickstarted a thriving community of 300+ members, averaging 50-200 attendees per event, with under AED 20,000 p.a.

Treasurer (Founding Team) | Code Paris – NYU Paris, France — Jan 2025 – May 2025

- Designed educational frameworks and held workshops on NumPy, AI models, and advanced programming concepts

Team Leader of 5 NYUAD undergraduates | **Bloomberg Trading Challenge** — Sep 2023 – Nov 2023

- Globally ranked 243, with net profit of 5% in 30 days over a diverse 1 million virtual USD portfolio